

Sai Balusu

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Education

Virginia Tech

Masters of Science in Mechanical Engineering | GPA:3.7

August 2024 – May 2026

Virginia Tech

Bachelor of Science in Mechanical Engineering, Robotics and Mechatronics | GPA:3.6

August 2021 – May 2024

Skills

Software: AutoCAD, Autodesk Inventor, Autodesk Fusion 360, SolidWorks, ANSYS, NX, Lifecycle Visualization, Teamcenter, Arduino, Simulink, LTspice, SPICE, KiCAD (PCB)

Programming & Technical: Python, C, C++, ROS, Java, MATLAB

Professional Experience

Soft Materials and Structures Lab

Graduate Research Assistant

August 2024 – Present

- Developed a fully autonomous grasping system on a **Franka FR3** arm, integrating octopus-sucker adhesion, a **Gemini vision pipeline**, and a **hybrid force-vision controller** to mitigate soft-sensor hysteresis and creep; raised fragile-produce grasp success from **0%** (joint-torque baseline) to **100%**.
- Engineered a **seven-layer soft capacitive skin** from a custom **Liquid Metal (SIS-Ag-EGaIn) composite**, **Ecoflex**, and a plasma-treated PTFE electret; achieving a **20 mN** detection threshold, **>27 dB** SNR, and capacitance jitter **< 0.09 pF**.

FASER Lab

Undergraduate Research Assistant

September 2023 – May 2024

- Led end-to-end design of a **Stewart platform** for **NASA's Borg Project**; achieved a **0.5m** range of motion through iterative **SolidWorks** prototyping and fabrication.
- Developed a **Python and ROS-based** robotic strut assembly system for NASA truss research; utilized simulation and experimental validation to minimize assembly errors and enhance mission safety.

General Motors

Product Design Engineering Intern

May 2023 – August 2023

- Reconstructed Chevy truck door water deflectors using **NX sheet body modeling**; simplified component architecture to meet production-quality standards and improve manufacturability.
- Standardized **NX CAD templates** for impact bars and reinforcements, reducing base-part creation time from 3 hours to **20 minutes** and driving department-wide adoption.
- Evaluated 4 door components for **weld compliance**; resolved manufacturability risks to ensure successful production release.

Soft Materials and Structures Lab

Undergraduate Research Assistant

January 2023 – May 2023

- Synthesized **LM-CNT-SIS composite** fabrication methods to prevent conductivity degradation; validated a **segregated network** that maintains electrical performance under high mechanical strain.

General Motors

Product Design Engineering Intern

May 2022 – August 2022

- Optimized **Equinox EV** seat packaging and carpet wrap geometry to **±0.02 cm** production tolerances, securing prototype approval and manufacturing feasibility.
- Validated 7+ components and assemblies via **NX and Lifecycle Visualization**; identified and corrected a **critical design error**, preventing potential production failure.

Solar Car at VT

Founding Member, Ex-Vice President, Mechanical Team

August 2021 - May 2024

- Co-founded the university's Solar Car team; spearheaded engineering workflows and organizational growth, leading to a **12th-place finish** at the **2024 American Solar Challenge (ASC)** and qualifying for **FSGP**.
- Led the suspension system from concept to production; utilized **SolidWorks** for design and secured critical sponsorships to meet all competition safety and performance regulations.
- Conducted comprehensive FEA using **ANSYS** to optimize suspension geometry; developed standardized simulation files that serve as the technical foundation for future vehicle iterations.
- Directed the end-to-end fabrication of suspension components; performed hands-on machining and assembly to ensure the vehicle successfully passed all **technical scrutineering** phases.

Personal Project

Humanoid Leg

Personal Project

December 2025 – Present

- Designed and fabricated a **4-DOF humanoid leg** using **SolidWorks** and 3D printing; optimized for high-agility maneuvers through iterative actuator selection and mechanical validation.
- Integrated **BLDC motors** with **ODrive S1 FOC controllers** via **CAN communication**; tuned low-level position, velocity, and torque control loops for precise joint-level trajectory tracking.
- Engineered a custom **cycloidal gearbox** with an **8:1 reduction ratio**, achieving **70% torque efficiency** and high back-drivability to enhance control authority across the leg's workspace.